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EXPANDABLE INTERVERTEBRAL IMPLANT

Abstract

An expandable intervertebral implant may include a superior endplate configured to engage a superior vertebra, and an inferior endplate configured to engage an inferior vertebra, wherein the inferior endplate may be spaced apart from the superior endplate by a gap. The expandable intervertebral implant may further include a first slider moveable along a first direction to urge linear expansion of the gap, and one or more cams each coupled to the first slider. The one or more cams may be configured to urge angular expansion of the gap in response to at least one of: rotation of the one or more cams about a rotation axis, and motion of the one or more cams along the first direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present applications claims the benefit of U.S. Provisional Ser. No. 63/557,480, filed on Feb. 24, 2024 and entitled EXPANDABLE INTERVERTEBRAL IMPLANT, which is incorporated by reference as though set forth herein in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to systems and methods for use in orthopedic surgery. More specifically, the present disclosure relates to expandable intervertebral implant systems and surgical methods.

BACKGROUND

[0003] Intervertebral disc degeneration and associated spinal disorders present significant clinical challenges, often leading to chronic pain and reduced mobility in affected individuals. Current surgical interventions primarily involve spinal fusion techniques or the use of fixed-size intervertebral implants. While these approaches aim to stabilize the spine and alleviate symptoms, they often limit natural spinal motion and may contribute to complications such as adjacent segment disease.

[0004] Fixed-size intervertebral implants currently available on the market vary in design and material composition but share common limitations. These implants require precise pre-operative sizing and may necessitate intraoperative adjustments to ensure proper fit within the disc space. Surgeons may also face challenges in adapting these implants to individual patient anatomy and achieving optimal alignment post-implantation. Moreover, the complexity of implantation procedures can prolong surgery times and recovery periods for patients.

[0005] Recognizing these challenges, there is a growing demand for intervertebral implants that offer greater versatility and precision in surgical application. An ideal solution would provide adjustable sizing capabilities intraoperatively, facilitating tailored fitment to varying disc spaces and spinal geometries. Such advancements could potentially improve surgical outcomes, minimize recovery times, and reduce the incidence of postoperative complications associated with current implant technologies.

SUMMARY

[0006] The various systems and methods of the present disclosure have been developed in response to the present state of the art, and in particular, in response to the problems and needs in the art that have not yet been fully solved by currently available expandable intervertebral implant systems and surgical methods.

[0007] In some embodiments, an expandable intervertebral implant may include a superior endplate configured to engage a superior vertebra, and an inferior endplate configured to engage an inferior vertebra, wherein the inferior endplate may be spaced apart from the superior endplate by a gap. The expandable intervertebral implant may further include a first slider moveable along a first direction to urge linear expansion of the gap, and one or more cams each coupled to the first slider. The one or more cams may be configured to urge angular expansion of the gap in response to at least one of: rotation of the one or more cams about a rotation axis, and motion of the one or more cams along the first direction.

[0008] In the expandable intervertebral implant of any preceding paragraph, the expandable intervertebral implant may further include a second slider threadably coupled to the first slider. The second slider may be moveable along the first direction, the second slider may be moveable independent of the first slider, and movement of the second slider may result in at least one of: rotation of the one or more cams about the rotation axis, and motion of the one or more cams along the first direction.

[0009] In the expandable intervertebral implant of any preceding paragraph, the expandable intervertebral implant may further include a first screw, extending along an anterior-posterior direction, configured to rotatably engage the first slider. The first direction may include the anterior-posterior direction, and rotation of the first screw may result in movement of the first slider in the anterior-posterior direction.

[0010] In the expandable intervertebral implant of any preceding paragraph, the first screw may be captive relative to the second slider.

[0011] In the expandable intervertebral implant of any preceding paragraph, the expandable intervertebral implant may further include a second screw, extending along the anterior-posterior direction, configured to rotatably engage the second slider. The second screw may be configured to rotate to urge movement of the second slider in the anterior-posterior direction.

[0012] In the expandable intervertebral implant of any preceding paragraph, the second screw may be captive relative to the superior endplate.

[0013] In the expandable intervertebral implant of any preceding paragraph, the one or more cams may be configured to be rotatable and translatable relative to the superior endplate and the inferior endplate.

[0014] In some embodiments, an expandable intervertebral implant may include a superior endplate configured to engage a superior vertebra, and an inferior endplate configured to engage an inferior vertebra, wherein the inferior endplate may be spaced apart from the superior endplate by a gap. The expandable intervertebral implant may further include one or more cams configured to be translatable relative to the superior endplate and the inferior endplate, a first screw extending along an anterior-posterior direction, and a first slider configured to threadably receive the first screw. The first screw may be configured to rotate to urge translation of the first slider in the anterior-posterior direction, the one or more cams may be coupled to the first slider such that translation of the first slider in the anterior-posterior direction results in translation of the one or more cams, and the one or more cams may be configured to urge angular expansion of the gap in response to translation of the one or more cams.

[0015] In the expandable intervertebral implant of any preceding paragraph, the expandable intervertebral implant may further include a second slider threadably coupled to the first slider and a second screw extending along the anterior-posterior direction. The second slider may be moveable along the anterior-posterior direction, the second screw may be configured to rotate to urge movement of the second slider in the anterior-posterior direction, and movement of the second slider may urge linear expansion of the gap.

[0016] In the expandable intervertebral implant of any preceding paragraph, the first slider may be moveable independent of the second slider.

[0017] In the expandable intervertebral implant of any preceding paragraph, movement of the second slider may result in comparable movement of the first slider and movement of the first slider may not result in movement of the second slider.

[0018] In the expandable intervertebral implant of any preceding paragraph, the one or more cams may be further configured to be rotatable relative to the superior endplate and the inferior endplate, wherein translation of the first slider in the anterior-posterior direction may result in rotation of the one or more cams, and the one or more cams may be configured to urge angular expansion of the gap in response to rotation of the one or more cams.

[0019] In the expandable intervertebral implant of any preceding paragraph, the expandable intervertebral implant may further include an anterior height having a first distance, in a superior-inferior plane, between a superior anterior portion and an inferior anterior portion, and a posterior height having a second distance, in the superior-inferior plane, between a superior posterior portion and an inferior posterior portion. In response to rotation of the one or more cams about a medial-lateral axis, the one or more cams may be configured to urge angular expansion of the gap by increasing the anterior height more than the posterior height.

[0020] In the expandable intervertebral implant of any preceding paragraph, the one or more cams may be received within one or more cam slots, and the one or more cam slots may be configured to urge rotation of the one or more cams relative to the first slider in response to translation of the one or more cams within the one or more cam slots.

[0021] In some embodiments, an expandable intervertebral implant may include a superior endplate configured to engage a superior vertebra, and an inferior endplate configured to engage an inferior vertebra, wherein the inferior endplate may be spaced apart from the superior endplate by a gap such that the expandable intervertebral implant may include, an anterior height having a first distance, in a superior-inferior plane, between a superior anterior portion of the superior endplate and an inferior anterior portion of the inferior endplate, and a posterior height having a second distance, in the superior-inferior plane, between a superior posterior portion and an inferior posterior portion. The expandable intervertebral implant may further include a first cam configured to rotate about a medial-lateral axis such that the gap may expand linearly and angularly by increasing the posterior height, and increasing the anterior height more than the posterior height.

[0022] In the expandable intervertebral implant of any preceding paragraph, the expandable intervertebral implant may further include a first slider coupled to the first cam. The first slider may be moveable along an anterior-posterior direction, and movement of the first slider may result in at least one of: rotation of the first cam about the medial-lateral axis, and motion of the first cam along the anterior-posterior direction.

[0023] In the expandable intervertebral implant of any preceding paragraph, the expandable intervertebral implant may further include a first screw, extending along the anterior-posterior direction, configured to rotatably engage the first slider, wherein the first screw may be configured to rotate to urge movement of the first slider in the anterior-posterior direction.

[0024] In the expandable intervertebral implant of any preceding paragraph, the first cam may be translatable in an anterior-posterior direction, and, in response to translation of the first cam, the first cam may be configured to urge angular expansion of the gap by increasing the anterior height more than the posterior height.

[0025] In the expandable intervertebral implant of any preceding paragraph, the first cam may be received within a cam slot, and the cam slot may be configured to urge rotation of the first cam relative to the cam slot in response to translation of the first cam within the cam slot.

[0026] In the expandable intervertebral implant of any preceding paragraph, the expandable intervertebral implant may further include a second slider moveable along a first direction to urge linear expansion of the gap.

[0027] These and other features and advantages of the present disclosure will become more fully apparent from the following description and appended claims or may be learned by the practice of the implants, systems, and methods set forth hereinafter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] Exemplary embodiments of the disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings depict only exemplary embodiments and are, therefore, not to be considered limiting of the scope of the appended claims, the exemplary embodiments of the disclosure will be described with additional specificity and detail through use of the accompanying drawings.

[0029] FIG. 1 is a perspective view of an expandable intervertebral implant in a collapsed configuration according to an embodiment of the present disclosure.

[0030] FIG. 2 is a perspective view of the expandable intervertebral implant of FIG. 1 in an

expanded configuration.

[0031] FIGS. 3A, 3B, 3C and 3D are perspective, top, front and side views of the expandable intervertebral implant of FIG. 1 in a collapsed configuration.

[0032] FIGS. 4A, 4B, 4C and 4D are perspective, top, front and side views of the expandable intervertebral implant of FIG. 1 in an expanded configuration.

[0033] FIG. 5 is a perspective exploded view of the expandable intervertebral implant of FIG. 1.

[0034] FIGS. 6A, 6B, 6C, 6D, and 6E are perspective, bottom perspective, top, front and side views of a superior endplate of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0035] FIGS. 7A, 7B, 7C, 7D, and 7E are perspective, bottom perspective, top, front and side views of an inferior endplate of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0036] FIGS. 8A, 8B, 8C, and 8D are perspective, left side, right side, and front views of a height saddle of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0037] FIGS. 9A, 9B, 9C, and 9D are perspective, top, front, and side views of a height slider of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0038] FIGS. 10A, 10B, and 10C are perspective, front, and side views of a height screw of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0039] FIGS. 11A, 11B, and 11C are perspective, front, and side views of an angulation screw of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0040] FIGS. 12A, 12B, 12C, and 12D are perspective, top, front, and side views of an angulation slider of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0041] FIGS. 13A, 13B, and 13C are perspective, front, and side views of an angulation cam of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0042] FIGS. 14A, 14B, and 14C are perspective, front, and side views of an angulation screw clip of the expandable intervertebral implant of FIG. 1 according to an embodiment of the present disclosure.

[0043] FIG. 15 is a perspective view of the expandable intervertebral implant of FIG. 1 in a collapsed configuration.

[0044] FIG. 16 is a perspective view of the expandable intervertebral implant of FIG. 1 assembled with a plate and fasteners according to an embodiment of the present disclosure.

[0045] FIGS. 17A, 17B and 17C are a front perspective, side, and top perspective view of the expandable intervertebral implant of FIG. 1 showing the steps to transition from a collapsed configuration to an intermediate expanded configuration.

[0046] FIGS. 18A, 18B and 18C are a front perspective, top perspective, and side view of the expandable intervertebral implant of FIG. 1 showing the steps to transition from a collapsed configuration to an expanded configuration.

[0047] FIGS. 19A, 19B, 19C, and 19D are section side views of the expandable intervertebral implant of FIG. 1 showing the engagement of the angulation cam for various states of expansion according to an embodiment of the present disclosure.

[0048] It is to be understood that the drawings are for purposes of illustrating the concepts of the present disclosure and may not be drawn to scale. Furthermore, the drawings illustrate exemplary embodiments and do not represent limitations to the scope of the present disclosure.

DETAILED DESCRIPTION

[0049] Exemplary embodiments of the present disclosure will be best understood by reference to the drawings, wherein like parts are designated by like numerals throughout. It will be readily

understood that the components of the present disclosure, as generally described and illustrated in the drawings, could be arranged, and designed in a wide variety of different configurations. Thus, the following more detailed description of the embodiments of the devices, systems, and methods, as represented in the drawings, is not intended to limit the scope of the present disclosure but is merely representative of exemplary embodiments of the present disclosure.

[0050] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. While the various aspects of the embodiments are presented in the drawings, the drawings are not necessarily drawn to scale unless specifically indicated.

[0051] Standard medical planes of reference and descriptive terminology are employed in this specification. While these terms are commonly used to refer to the human body, certain terms are applicable to physical objects in general.

[0052] A standard system of three mutually perpendicular reference planes is employed. A sagittal plane divides a body into right and left portions. A coronal plane divides a body into anterior and posterior portions. A transverse plane divides a body into superior and inferior portions. A mid-sagittal, mid-coronal, or mid-transverse plane divides a body into equal portions, which may be bilaterally symmetric. The intersection of the sagittal and coronal planes defines a superior-inferior or cephalad-caudal axis. The intersection of the sagittal and transverse planes defines an anterior-posterior axis. The intersection of the coronal and transverse planes defines a medial-lateral axis. The superior-inferior or cephalad-caudal axis, the anterior-posterior axis, and the medial-lateral axis are mutually perpendicular.

[0053] Anterior means toward the front of a body. Posterior means toward the back of a body. Superior or cephalad means toward the head. Inferior or caudal means toward the feet or tail. Medial means toward the midline of a body, particularly toward a plane of bilateral symmetry of the body. Lateral means away from the midline of a body or away from a plane of bilateral symmetry of the body. Axial means toward a central axis of a body. Abaxial means away from a central axis of a body. Ipsilateral means on the same side of the body. Contralateral means on the opposite side of the body. Proximal means toward the trunk of the body. Proximal may also mean toward a user or operator. Distal means away from the trunk. Distal may also mean away from a user or operator. Dorsal means toward the top of the foot. Plantar means toward the sole of the foot. Varus means deviation of the distal part of the leg below the knee inward, resulting in a bowlegged appearance. Valgus means deviation of the distal part of the leg below the knee outward, resulting in a knock-kneed appearance.

[0054] The present disclosure relates to intervertebral implant devices, systems, and methods. Those skilled in the art will recognize that the following description is merely illustrative of the principles of the technology, which may be applied in various ways to provide many alternative embodiments. The present disclosure illustrates devices for an anterior lumbar interbody fusion (ALIF) for the purposes of illustrating the concepts of the present design. However, it will be understood that other variations and uses are contemplated including, but not limited to, devices for a posterior lumbar interbody fusion (PLIF), devices for transforaminal lumbar interbody fusion (TLIF), devices for interbody fusion of the thoracic spine, devices for interbody fusion of the cervical spine, etc.

[0055] FIG. 1 is a perspective view of an expandable intervertebral implant **2000** in a collapsed configuration according to an embodiment of the present disclosure. FIG. 2 is a perspective view of the expandable intervertebral implant of FIG. 1 in an expanded configuration. In some embodiments, the expandable intervertebral implant **2000** may be configured to be implanted within a spinal disc space following a discectomy procedure during an anterior lumbar interbody fusion (ALIF) procedure. The expandable intervertebral implant **2000** may be implanted between a superior vertebra and an inferior vertebra to support fusion of the vertebral bodies. The expandable

intervertebral implant **2000** may be configured to include a superior endplate **100** and an inferior endplate **200**. The superior endplate **100** may include a superior anterior portion **130** and a superior posterior portion **140**. The inferior endplate **200** may include an inferior anterior portion **230** and an inferior posterior portion **240**.

[0056] The expandable intervertebral implant **2000** may include a superior endplate **100** configured to engage a superior vertebra; an inferior endplate **200** configured to engage an inferior vertebra and spaced apart from the superior endplate **100** by a gap, a height slider **400** moveable along an anterior-posterior direction to urge linear expansion of the gap, and an angulation slider **700** moveable along an anterior-posterior direction, to urge angular expansion of the gap. The linear expansion may be activated by engaging one or more pairs of angled surfaces. The angular expansion may be activated by a rotation and/or translation of a cam mechanism.

[0057] FIGS. **3A**, **3B**, **3C** and **3D** are perspective, top, front and side views of the expandable intervertebral implant **2000** in a collapsed configuration. The expandable intervertebral implant **2000** may have an anterior height **2010** and a posterior height **2020**. The anterior height **2010** may be the distance, in a superior-inferior plane, between the superior anterior portion **130** and the inferior anterior portion **230**. The posterior height **2020** may be the distance, in a superior-inferior plane, between the superior posterior portion **140** and the inferior posterior portion **240**.

[0058] The expandable intervertebral implant **2000** may be configured to include a collapsed (un-expanded) configuration and one or more expanded configurations. In the collapsed configuration, the anterior height **2010** may be at a minimum of an anterior travel range and the posterior height **2020** may be at a minimum of a posterior travel range. Additionally, the expandable intervertebral implant **2000** may be configured for multiple size variations whereby the anterior height **2010** may range from 8 mm to 20 mm in the collapsed configuration. The expandable intervertebral implant **2000** may include an implant angle **2030**. The expandable intervertebral implant **2000** may be configured for multiple implant angle **2030** variations ranging from 0° to 20° in the collapsed configuration.

[0059] The expandable intervertebral implant **2000** may be configured to be placed into a disc space in a collapsed configuration and expanded in-situ to restore an anatomical intervertebral disc height and/or an angle between the superior vertebra and the inferior vertebra. The expandable intervertebral implant **2000** may be configured to expand in height so that the anterior height **2010** and the posterior height **2020** increase at generally the same rate as each other during the expansion process. The expandable intervertebral implant **2000** may further be configured to expand at an angle whereby the anterior height **2010** is increased and the posterior height **2020** remains generally unchanged.

[0060] In some embodiments, the expandable intervertebral implant **2000** may be configured so that a right side of the expandable intervertebral implant **2000** and a left side of the expandable intervertebral implant **2000** may be expanded independently of each other in a medial-lateral plane. Additionally, or alternatively, the expandable intervertebral implant **2000** may be configured so that the height expansion of the expandable intervertebral implant **2000** may be independent of the angular expansion of the expandable intervertebral implant **2000**. Additionally, or alternatively, the actuation mechanism for angular expansion may be independent of the actuation mechanism for linear expansion.

[0061] The expandable intervertebral implant **2000** may be configured so that the height expansion, whereby the anterior height **2010** and the posterior height **2020** increase at generally the same rate as each other during the expansion process, may be actuated by a first means, and the angular expansion, whereby the anterior height **2010** is increased and the posterior height remains generally unchanged, may be actuated by a second means. In some embodiments the first means of actuation and the second means of actuation are distinct and may be engaged independently of each other. The first means of actuation and the second means of actuation may be engaged in any sequence. The first means of actuation may include a first threaded mechanism. The second means of

actuation may include a second threaded mechanism.

[0062] FIGS. 4A, 4B, 4C and 4D are perspective, top, front and side views of the expandable intervertebral implant **2000** in an expanded configuration. The implant angle **2030** of the expandable intervertebral implant **2000** may be increased through angular expansion to generally match the lordotic angle of a superior vertebra and an inferior vertebra. The expandable intervertebral implant **2000** may be configured wherein angular expansion increases the anterior height **2010** without generally increasing the posterior height **2020**. In some embodiments, the expandable intervertebral implant **2000** may be configured for a posterior lumbar interbody fusion (PLIF) procedure, wherein angular expansion may increase the posterior height **2020** without generally increasing the anterior height **2010**. The range of angular expansion may be between 0° and 15°.

[0063] The height of the expandable intervertebral implant **2000** may be increased through height expansion. The expandable intervertebral implant **2000** may be configured wherein height expansion increases both the anterior height **2010** and the posterior height **2020** by generally equal amounts. The range of height expansion may be between 0 mm and 8 mm.

[0064] FIG. 5 is a perspective exploded view of the expandable intervertebral implant **2000**. The expandable intervertebral implant **2000** may include a superior endplate **100**, an inferior endplate **200**, a height saddle **300**, a mirror image height saddle **300'**, a height slider **400**, a height screw **500**, an angulation screw **600**, an angulation slider **700**, one or more angulation cams **800**, an angulation screw clip **900**, one or more cam pins **1000**, one or more slider pins **1100**, and one or more endplate pins **1200**.

[0065] FIGS. 6A, 6B, 6C, 6D, and 6E are perspective, bottom perspective, top, front and side views of a superior endplate **100** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The superior endplate **100** may be configured to engage a superior vertebra. The superior endplate **100** may include a central aperture **110**, and a plurality of peripheral apertures **117** to allow bone growth between a superior vertebra and an inferior vertebra through the expandable intervertebral implant **2000**.

[0066] The superior endplate **100** may further include a plurality of side apertures **115**. The superior endplate **100** may be configured to receive bone graft material within the central aperture **110** to facilitate bone ingrowth.

[0067] The superior endplate **100** may also include a pocket **120** and an opening **125**. The pocket **120** and the opening **125** may be located at the posterior portion of the superior endplate **100**. The pocket **120** and the opening **125** may be configured to rotatably receive the height screw **500**.

[0068] The superior endplate **100** may further include one or more pin apertures **150** configured to receive one or more endplate pins **1200**. The one or more endplate pins **1200** may also be received within one or more pin apertures **250** of the inferior endplate **200**. The one or more endplate pins **1200** may allow rotation of the superior endplate **100** relative to the inferior end plate whereby the axis of rotation may be colinear with a longitudinal axis of the one or more endplate pins **1200**. Additionally, the one or more endplate pins **1200** may be configured to translate along the length of the one or more pin apertures **250** as the height of the expandable intervertebral implant **2000** is increased.

[0069] The superior endplate **100** may further include one or more fastener channels **160** configured to allow passage of a fastener **1300** (as shown in FIG. 16) through the anterior portion of the expandable intervertebral implant **2000** and into an inferior vertebra. The superior endplate **100** may also include one or more fins **170** located on the superior anterior portion **130**. The one or more fins **170** may be configured to engage the superior vertebra and may prevent expulsion, subsidence and/or migration of the expandable intervertebral implant **2000** from an intervertebral space. Additionally, or alternatively, the one or more fins **170** may be located on the superior posterior portion **140**. Additionally, or alternatively, the one or more fins **170** may be located on a superior surface of the superior endplate **100**.

[0070] The superior endplate **100** may include a first channel **180** and a second channel **185**. The first channel **180** and the second channel **185** may be configured to slidably receive the height slider **400** and/or the angulation slider **700**. The first channel **180** and the second channel **185** may each be configured with an undercut portion along one or more sides along the length of the first channel **180** and/or the second channel **185**. For example, the first channel **180** and the second channel **185** may be configured with a partial dovetail profile and/or a full dovetail profile.

[0071] The first channel **180** and the second channel **185** may be configured so that the height slider **400** and/or the angulation slider **700** are slidably captive relative to the first channel **180** and/or the second channel **185**. The height slider **400** and/or the angulation slider **700** may move with the superior endplate **100** during linear expansion and/or angular expansion of the expandable intervertebral implant **2000**. Additionally, or alternatively, the height slider **400** and/or the angulation slider **700** may move within the first channel **180** and/or the second channel **185** during linear expansion and/or angular expansion of the expandable intervertebral implant **2000**.

[0072] As used herein, “captive relative to” refers to a first component that is directly coupled with a second component such that, during movement of either component, the two components remain directly coupled. Additionally, the two components may rotate relative to each other and/or translate relative to each other while remaining directly coupled.

[0073] The superior endplate **100** may have a plurality of aperture side walls **190** at the edges of the central aperture **110**. The plurality of aperture side walls **190** may be generally parallel to the first channel **180** and the second channel **185**. The superior endplate **100** may also include a boss width **195**. The boss width **195** may be greater than the pocket **120** and the opening **125**.

[0074] FIGS. 7A, 7B, 7C, 7D, and 7E are perspective, bottom perspective, top, front and side views of an inferior endplate **200** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The inferior endplate **200** may be configured to engage an inferior vertebra. The inferior endplate **200** may include a central aperture **210**, and a plurality of peripheral apertures **217** to allow bone growth between a superior vertebra and an inferior vertebra through the expandable intervertebral implant **2000**. The inferior endplate **200** may be configured to receive bone graft material within the central aperture **210** to facilitate bone ingrowth.

[0075] The inferior endplate **200** may include one or more first angled portions **220** and one or more second angled portions **225**. The one or more first angled portions **220** may be configured to slidably engage one or more third angled portions **320** of the height saddle **300**. The one or more second angled portions **225** may be configured to slidably engage one or more fourth angled portions **325** of the height saddle **300**. The one or more first angled portions **220** and the one or more second angled portions **225** may be configured so that a force applied in a posterior direction from the one or more third angled portions **320** and/or one or more fourth angled portions **325** may result in translation of the superior endplate **100**.

[0076] The one or more first angled portions **220** and the one or more second angled portions **225** may each include one or more side grooves **290**. The one or more side grooves **290** may be configured as an undercut portion along one or more sides along the length of the one or more first angled portions **220** and/or the one or more second angled portions **225**. For example, the one or more side grooves **290** may be configured with a partial dovetail profile and/or a full dovetail profile.

[0077] The first angled portion **220** and the second angled portion **225** may be included in one or more saddle channels **260**. The one or more saddle channels **260** may slidably receive one or more height saddles **300** and/or one or more height saddles **300'** and may guide movement of the one or more height saddles **300** and/or one or more height saddles **300'** in a generally anterior-posterior direction.

[0078] The inferior endplate **200** may also include one or more fins **270** located on the inferior anterior portion **230**. The one or more fins **270** may be configured to engage the superior vertebra and may prevent expulsion, subsidence and/or migration of the expandable intervertebral implant

2000 from an intervertebral space. Additionally, or alternatively, the one or more fins **270** may be located on the inferior posterior portion **240**. Additionally, or alternatively, the one or more fins **270** may be located on an inferior surface of the inferior endplate **200**.

[0079] The inferior endplate **200** may further include a posterior opening **280** and an anterior opening **285**. The posterior opening **280** may be configured so that the height screw **500** does not impact the inferior endplate **200** over the entire range of travel of the height screw **500** when the expandable intervertebral implant **2000** is in the collapsed configuration, the one or more expanded configurations and/or transitioning between the collapsed configuration and the one or more expanded configurations. The anterior opening **285** may be configured to allow a driver to engage the height screw **500** and/or the angulation screw **600** through an anterior portion of the inferior endplate **200** when the expandable intervertebral implant **2000** is in the collapsed configuration, the one or more expanded configurations and/or transitioning between the collapsed configuration and the one or more expanded configurations.

[0080] The inferior endplate **200** may further include one or more fastener apertures **295** configured to allow passage of a fastener **1300** (as shown in FIG. **16**) through the anterior portion of the expandable intervertebral implant **2000** and into an inferior vertebra.

[0081] FIGS. **8A**, **8B**, **8C**, and **8D** are perspective, left side, right side, and front views of a height saddle **300** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The expandable intervertebral implant **2000** may include a height saddle **300** and a height saddle **300'** wherein the height saddle **300'** may be a complete mirror image of the height saddle **300**. All features subsequently described for the height saddle **300** may also be present on the height saddle **300'**.

[0082] The height saddle **300** may be configured to translate anterior-posterior motion into inferior-superior motion. More specifically, as the height saddle **300** is urged in a posterior direction by the height slider **400**, which may be actuated by the height screw **500**, the height saddle **300** may slidably engage one or more angled surfaces of the inferior endplate **200** resulting in movement of the height saddle **300** in a superior direction thereby urging the superior endplate **100** in a superior direction.

[0083] The height saddle **300** may include one or more third angled portions **320** and one or more fourth angled portions **325**. The one or more third angled portions **320** may be configured to slidably engage one or more first angled portions **220** of the inferior endplate **200**. The one or more fourth angled portions **325** may be configured to slidably engage one or more second angled portions **225** of the inferior endplate **200**. The one or more third angled portions **320** and/or the one or more fourth angled portions **325** may be configured so that a force applied in a posterior direction from height slider **400** may result in translation of the superior endplate **100**.

[0084] The one or more third angled portions **320** may include one or more first engagement portions **330**. The one or more fourth angled portions **325** may include one or more second engagement portions **335**. The one or more first engagement portions **330** and/or the one or more second engagement portions **335** may be configured to slidably engage the one or more side grooves **290** of the inferior endplate **200**.

[0085] The height saddle **300** may have a saddle width **340**. The saddle width **340** may be less than or equal to the width of the one or more saddle channels **260** of the inferior endplate **200**. The one or more first engagement portions **330** and/or the one or more second engagement portions **335** may be configured so that the height saddle **300** may be slidably captive relative to the one or more saddle channels **260** of the inferior endplate **200**.

[0086] The height saddle **300** may include a cam slot **310** and a pin aperture **315**. The cam slot **310** may be configured to receive a cam pin **1000**. The pin aperture **315** may be configured to receive a slider pin **1100**. The height saddle **300** may also include a first cam engagement surface **350**, a second cam engagement surface **360**, and a third cam engagement surface **370**. The cam slot **310** in conjunction with the cam pin **1000** may guide the translation and/or rotation of an angulation cam

800 along the first cam engagement surface **350**, the second cam engagement surface **360**, and the third cam engagement surface **370**.

[0087] FIGS. **9A**, **9B**, **9C**, and **9D** are perspective, top, front, and side views of a height slider **400** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The height slider **400** may be configured to slidably engage the superior endplate **100**. The height slider **400** may further be configured to translate rotational motion of the height screw **500** into linear motion of the height slider **400** with respect to the superior endplate **100**. Movement of the height slider **400** may result in comparable movement of the angulation slider **700**.

[0088] The height slider **400** may include a central portion **410**, a first arm **412**, and a second arm **414**. The central portion **410** may include a first pocket **430**, a second pocket **435**, an inside diameter **445**, and an internal thread **440**. The internal thread **440** may be configured to threadably receive the height screw **500**. The internal thread **440** may include a left-hand thread having a thread pitch and a pitch diameter compatible with the external thread **510** of the height screw **500**.

[0089] The inside diameter **445** may be configured to rotatably receive the angulation screw **600**. The first pocket **430** and the second pocket **435** may be configured to receive the angulation screw clip **900**. More specifically, the first pocket **430** may receive the one or more legs **910** and the first locating feature **920** of the angulation screw clip **900**. The second pocket **435** may receive the second locating feature **930** of the angulation screw clip **900**. The height slider **400**, the angulation screw **600**, and the angulation screw clip **900** may be configured so that the angulation screw **600** is rotatably captive relative to the inside diameter **445**.

[0090] The first arm **412** may include a first engagement portion **420** and the second arm **414** may include a second engagement portion **425**. The first engagement portion **420** may be configured to slidably engage the first channel **180** of the superior endplate **100**. The second engagement portion **425** may be configured to slidably engage the second channel **185** of the superior endplate **100**. The first engagement portion **420** and the second engagement portion **425** may each be configured with an undercut portion configured to slidably engage a complimentary undercut portion along one or more sides along the length of the first channel **180** and/or the second channel **185**. For example, the first channel **180** and the second channel **185** may be configured with a partial female dovetail profile or a full female dovetail profile and the first engagement portion **420** and the second engagement portion **425** may each include a complimentary partial male dovetail profile or a full male dovetail profile.

[0091] The first arm **412** and the second arm **414** may each include a pin aperture **450**. The pin aperture **450** may be configured to receive a slider pin **1100**. The slider pin **1100** may allow rotation of the height slider **400** with respect to the height saddle **300** whereby the axis of rotation may be colinear with a longitudinal axis of the slider pin **1100** and the pin aperture **450**.

[0092] FIGS. **10A**, **10B**, and **10C** are perspective, front, and side views of a height screw **500** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The height screw **500** may be a component of an actuation mechanism configured to translate rotational motion into linear motion. The height screw **500** may include a head **520** and a neck **525**. The head **520** may be rotatably captive relative to the pocket **120** of the superior endplate **100** and the neck **525** may rotatably engage the opening **125** of the superior endplate **100**, thereby restricting linear motion of the height screw **500** with respect to the superior endplate **100**.

[0093] The height screw **500** may have an external thread **510** that may be configured to threadably engage the internal thread **440** of the height slider **400**. The external thread **510** may include a left-hand thread having a thread pitch and a pitch diameter compatible with the internal thread **440** of the height slider **400**.

[0094] The height screw **500** may also include a drive feature **530**. The drive feature **530** may have a hexalobe geometry or the like; this geometry may match that of a tip portion of a driver (not shown) that may impart torque to the height screw **500** through rotation of the driver. The drive feature **530** may have a hex, a hexalobe, a square or other non-circular profile known in the art. In

alternative embodiments, the drive feature **530** may have a different geometry or may be replaced by one or more positive features such as a boss or any other driver tip geometry known in the art. [0095] FIGS. **11A**, **11B**, and **11C** are perspective, front, and side views of an angulation screw **600** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The angulation screw **600** may be a component of an actuation mechanism configured to translate rotational motion into linear motion. The angulation screw **600** may include a head **620** and a neck **625**. The head **620** may be rotatably captive relative to the inside diameter **445** of the height slider **400** and the neck **625** may rotatably engage the first locating feature **920** of the angulation screw clip **900**, thereby restricting linear motion of the angulation screw **600** with respect to the height slider **400**.

[0096] The angulation screw **600** may have an external thread **610** that may be configured to threadably engage the internal thread **740** of the angulation slider **700**. The external thread **610** may include a right-hand thread having a thread pitch and a pitch diameter compatible with the internal thread **740** of the angulation slider **700**.

[0097] The angulation screw **600** may also include a drive feature **630**. The drive feature **630** may have a hexalobe geometry or the like; this geometry may match that of a tip portion of a driver (not shown) that may impart torque to the angulation screw **600** through rotation of the driver. The drive feature **630** may have a hex, a hexalobe, a square or other non-circular profile known in the art.

[0098] The angulation screw **600** may also include an internal diameter **640**. The internal diameter **640** may allow a driver to pass through the angulation screw **600** and engage the height screw **500**. The internal diameter **640** may further allow rotation of the driver while engaged with the height screw **500** thereby imparting a torque to the height screw **500**. The internal diameter **640** may be larger than a standard minor diameter of a hexalobe drive feature resulting in a modified hexalobe drive feature.

[0099] FIGS. **12A**, **12B**, **12C**, and **12D** are perspective, top, front, and side views of an angulation slider **700** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The angulation slider **700** may be configured to slidably engage the superior endplate **100**. The angulation slider **700** may further be configured to translate rotational motion of the angulation screw **600** into linear motion of the angulation slider **700** with respect to the height slider **400**. Movement of the angulation slider **700** may not result in movement of the height slider **400**.

[0100] The angulation slider **700** may include a central portion **710**, a first arm **712**, and a second arm **714**. The central portion **710** may include an internal thread **740** configured to threadably receive the angulation screw **600**. The internal thread **740** may include a right-hand thread having a thread pitch and a pitch diameter compatible with the external thread **610** of the angulation screw **600**.

[0101] The first arm **712** may include a first engagement portion **720** and the second arm **714** may include a second engagement portion **725**. The first engagement portion **720** may be configured to slidably engage the first channel **180** of the superior endplate **100**. The second engagement portion **725** may be configured to slidably engage the second channel **185** of the superior endplate **100**. The first engagement portion **720** and the second engagement portion **725** may each be configured with an undercut portion configured to slidably engage a complimentary undercut portion along one or more sides along the length of the first channel **180** and/or the second channel **185**. For example, the first channel **180** and the second channel **185** may be configured with a partial female dovetail profile or a full female dovetail profile and the first engagement portion **720** and the second engagement portion **725** may each include a complimentary partial male dovetail profile or a full male dovetail profile.

[0102] The first arm **712** and the second arm **714** may each include a pin aperture **750**. The pin aperture **750** may be configured to receive a cam pin **1000**. The cam pin **1000** may be received in the cam slot **310** of the height saddle **300**. Additionally, the cam pin **1000** in conjunction with the

cam slot **310** may guide the movement and/or rotation of an angulation cam **800** whereby the axis of rotation may be colinear with an axis of the cam pin **1000** and the pin aperture **750**.

[0103] The first arm **712** may further include a first cam pocket **760** and the second arm **714** may further include a second cam pocket **765**. The first cam pocket **760** and the second cam pocket **765** may each be configured to receive an angulation cam **800**. The pin aperture **750** may extend into each of the first arm **712** and the second arm **714** so that the cam pin **1000** may allow an angulation cam **800** to be rotatably captive relative to each of the first cam pocket **760** and the second cam pocket **765**.

[0104] The first arm **712** and the second arm **714** may each have a locating boss **770** and one or more cam locating surfaces **780**. The locating boss **770** may be configured so that, when the first engagement portion **720** is slidably received within the first channel **180** of the superior endplate **100**, the locating boss **770** is proximate the aperture side wall **190** of the superior endplate **100**, thereby helping keep the angulation slider **700** aligned within the first channel **180** and the second channel **185** resulting smooth motion of the angulation slider **700** with respect to the superior endplate **100**.

[0105] The one or more cam locating surfaces **780** may be configured to limit rotation of the one or more angulation cams **800**. The one or more cam locating surfaces **780** may engage a first locating surface **890** and/or a second locating surface **895** of the one or more angulation cams **800**.

[0106] FIGS. **13A**, **13B**, and **13C** are perspective, front, and side views of an angulation cam **800** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The one or more angulation cams **800** may be components of an actuation mechanism configured to translate rotational motion into angular motion. The angulation cam **800** may include a pin aperture **810** configured to receive a cam pin **1000**. The angulation cam **800** may be coupled to the angulation slider **700** by the cam pin **1000**.

[0107] The angulation cam **800** may further include an outside diameter **820**, a body portion **830**, and a locating portion **840**. The outside diameter **820** and the body portion **830** may be configured to allow the angulation cam **800** to be rotatably received within the first cam pocket **760** and/or the second cam pocket **765**. The pin aperture **810** may include a rotation axis **812** about which the angulation cam may rotate. The angulation cam **800** may be configured to be rotatable and translatable relative to the superior endplate **100** and the inferior endplate **200**.

[0108] The locating portion **840** may include a first cam surface **850**, a second cam surface **860**, a third cam surface **870**, a fourth cam surface **880**, a first locating surface **890**, and a second locating surface **895**. The first cam surface **850**, the second cam surface **860**, the third cam surface **870**, and/or the fourth cam surface **880** may engage the first cam engagement surface **350**, the second cam engagement surface **360**, and/or the third cam engagement surface **370** of the height saddle **300** as the expandable intervertebral implant **2000** transitions from a collapsed configuration to one or more expanded configurations. More specifically, the first cam surface **850**, the second cam surface **860**, the third cam surface **870**, and/or the fourth cam surface **880** may engage the first cam engagement surface **350**, the second cam engagement surface **360**, and/or the third cam engagement surface **370** of the height saddle **300** as the angulation screw **600** is rotated and the angulation slider translates from a first position to a second position.

[0109] FIGS. **14A**, **14B**, and **14C** are perspective, front, and side views of an angulation screw clip **900** of the expandable intervertebral implant **2000** according to an embodiment of the present disclosure. The angulation screw clip **900** may be configured to rotatably retain the angulation screw **600** within the height slider **400**. The angulation screw clip **900** may include one or more legs **910**, a first locating feature **920**, and a second locating feature **930**. The angulation screw clip **900** may be configured to be received within the first pocket **430** and the second pocket **435** of the height slider **400**. More specifically, the one or more legs **910** and the first locating feature **920** may be received within the first pocket **430** of the height slider **400**. The second locating feature **930** may be received within the second pocket **435** of the height slider **400**. The height slider **400**, the

angulation screw **600**, and the angulation screw clip **900** may be configured so that the angulation screw **600** is rotatably captive relative to the inside diameter **445** of the height slider **400**.

[0110] FIG. **15** is a perspective view of the expandable intervertebral implant **2000** in a collapsed configuration. FIG. **16** is a perspective view of the expandable intervertebral implant **2000** assembled with a plate **1400** and fasteners **1300** according to an embodiment of the present disclosure. The expandable intervertebral implant **2000** may include a plate receiving portion **2060**. The expandable intervertebral implant **2000** may use similar fasteners, similar instruments, and/or similar methods, including the plate **1400** and the fasteners **1300**, as described in U.S. Pat. No. 10,758,370 filed on Jul. 2, 2018, entitled “INTERBODY SPACER AND BONE PLATE ASSEMBLY”, the disclosure of which is incorporated herein by reference in its entirety.

[0111] FIGS. **17A**, **17B** and **17C** are a front perspective, side, and top perspective view of the expandable intervertebral implant **2000** showing the steps to transition from a collapsed configuration to an intermediate expanded configuration. A method for height expansion of the expandable intervertebral implant **2000** may include the following steps: engaging a first driver tool (not shown) with the drive feature **530** of the height screw **500** and rotating the first driver tool clock-wise. The clock-wise rotation of the height screw **500** may result in translation of the height slider **400** in a posterior direction. The height saddle **300** and the height saddle **300'** may be linked to the height slider **400** and may also translate in a posterior direction. The height saddle **300** and the height saddle **300'** may slidably engage the one or more first angled portions **220** and the one or more second angled portions **225** of the inferior endplate **200** resulting in translation of the height saddle **300** and the height saddle **300'**, as well as the height slider **400**, the angulation slider **700**, and the superior endplate **100**, in a superior direction.

[0112] FIGS. **18A**, **18B** and **18C** are a front perspective, top perspective, and side view of the expandable intervertebral implant **2000** showing the steps to transition from a collapsed configuration to an expanded configuration. A method for angular expansion of the expandable intervertebral implant **2000** may include the following steps: engaging a second driver tool (not shown) with the drive feature **630** of the angulation screw **600** and rotating the second driver tool clock-wise. The clock-wise rotation of the angulation screw **600** may result in translation of the angulation slider **700** in an anterior direction. The one or more angulation cams **800** may be linked to the angulation slider **700** whereby the anterior movement of the angulation slider **700** may result in the one or more angulation cams **800** translating and/or rotating while following the cam slot **310** of height saddle **300** and/or height saddle **300'**, resulting in movement of the angulation slider **700** in a superior direction, an increase in the anterior height **2010**, and an increase in the implant angle **2030**.

[0113] FIGS. **19A**, **19B**, **19C**, and **19D** are section side views of the expandable intervertebral implant **2000** showing the engagement of an angulation cam **800** for various states of expansion according to an embodiment of the present disclosure. FIG. **19A** shows the expandable intervertebral implant **2000** in a collapsed state wherein an endplate plane and a neutral plane are generally parallel. In the collapsed state the second cam surface **860** of the angulation cam **800** may be in contact with the first cam engagement surface **350** of the height saddle **300**.

[0114] FIG. **19B** shows the expandable intervertebral implant **2000** in a first expansion state, wherein an expansion angle **2040** between the endplate plane and the neutral plane may be greater than zero. One or more angulation cams **800** may translate and/or rotate along one or more cam slots **310** of the height saddle **300**, in response to clock-wise rotation of the angulation screw **600**, from the collapsed state to the first expansion state. In the first expansion state the third cam surface **870** of the angulation cam **800** may be in contact with the second cam engagement surface **360** of the height saddle **300**.

[0115] FIG. **19C** shows the expandable intervertebral implant **2000** in a second expansion state, wherein the expansion angle **2040** between the endplate plane and the neutral plane may be greater than the expansion angle **2040** in the first expansion state. One or more angulation cams **800** may

translate and/or rotate along one or more cam slots **310** of the height saddle **300**, in response to clock-wise rotation of the angulation screw **600**, from the first expansion state to the second expansion state. In the second expansion state the fourth cam surface **880** of the angulation cam **800** may be in contact with the third cam engagement surface **370** of the height saddle **300**.

[0116] FIG. **19D** shows the expandable intervertebral implant **2000** in a third expansion state, wherein the expansion angle **2040** between the endplate plane and the neutral plane may be greater than the expansion angle **2040** in the second expansion state. One or more angulation cams **800** may translate and/or rotate along one or more cam slots **310** of the height saddle **300**, in response to clock-wise rotation of the angulation screw **600**, from the second expansion state to the third expansion state. In the third expansion state the second cam surface **860** of the angulation cam **800** may be in contact with the third cam engagement surface **370** of the height saddle **300**.

[0117] An actuation device, external to an expandable intervertebral implant **2000**, may include one or more pairs of angled surfaces and/or one or more cam mechanisms configured to engage the expandable intervertebral implant **2000**. The actuation device may be configured to removably engage the expandable intervertebral implant **2000**. The actuation device may be configured to actuate height expansion and/or angular expansion of the expandable intervertebral implant **2000**. Additionally, the expandable intervertebral implant **2000** may be configured to retain the height expansion and/or angular expansion driven by the actuation device.

[0118] The expandable intervertebral implant **2000** may be manufactured from titanium alloy, titanium, stainless steel, cobalt chrome, PEEK, or other implant quality material with suitable mechanical properties. Additionally, or alternatively, one or more components of the expandable intervertebral implant **2000** may be manufactured through additive manufacturing methods. In some embodiments, one or more components of the expandable intervertebral implant **2000** may include a porous surface. The porous surface may be configured to promote bone in-growth. The porous surface may be configured as a trabecular structure configured to promote bone in-growth. Additionally, or alternatively, the porous surface may be configured to include cells that are configured to receive biologic material prior to implantation to promote bone in-growth after implantation. Additionally, or alternatively, the porous surface may be configured to include less metal mass implanted and more open areas for the bone to grow through the implant and create a more rigid fusion of the superior vertebra and inferior vertebra. In an embodiment, one or more components of the expandable intervertebral implant **2000** may be additively manufactured with integrated porous features configured for bone integration.

[0119] The expandable intervertebral implant **2000** may have flexible or compliant structures to perform or maintain expansion. The expandable intervertebral implant **2000** may have a single superior/inferior endplate component having a living hinge in place of a superior endplate **100** configured to be rotatably connected to an inferior endplate **200** and one or more endplate pins **1200**.

[0120] Reference throughout this specification to “an embodiment” or “the embodiment” means that a particular feature, structure or characteristic described in connection with that embodiment is included in at least one embodiment. Thus, the quoted phrases, or variations thereof, as recited throughout this specification are not necessarily all referring to the same embodiment.

[0121] Similarly, it should be appreciated that in the above description of embodiments, various features are sometimes grouped together in a single embodiment, Figure, or description thereof for the purpose of streamlining the disclosure. This method of disclosure, however, is not to be interpreted as reflecting an intention that any claim require more features than those expressly recited in that claim. Rather, as the following claims reflect, inventive aspects lie in a combination of fewer than all features of any single foregoing disclosed embodiment. Thus, the claims following this Detailed Description are hereby expressly incorporated into this Detailed Description, with each claim standing on its own as a separate embodiment. This disclosure includes all permutations of the independent claims with their dependent claims.

[0122] The phrases “generally parallel” and “generally perpendicular” refer to structures that are within 30° parallelism or perpendicularity relative to each other, respectively. Recitation in the claims of the term “first” with respect to a feature or element does not necessarily imply the existence of a second or additional such feature or element. Elements recited in means-plus-function format are intended to be construed in accordance with 35 U.S.C. § 112 Para. 6. It will be apparent to those having skill in the art that changes may be made to the details of the above-described embodiments without departing from the underlying principles of the disclosure.

[0123] While specific embodiments and applications of the present disclosure have been illustrated and described, it is to be understood that the disclosure is not limited to the precise configuration and components disclosed herein. Various modifications, changes, and variations which will be apparent to those skilled in the art may be made in the arrangement, operation, and details of the methods and systems of the present disclosure without departing from its spirit and scope.

Claims

1. An expandable intervertebral implant comprising: a superior endplate configured to engage a superior vertebra; an inferior endplate configured to engage an inferior vertebra, wherein the inferior endplate is spaced apart from the superior endplate by a gap; a first slider moveable along a first direction to urge linear expansion of the gap; and one or more cams each coupled to the first slider; wherein the one or more cams are configured to urge angular expansion of the gap in response to at least one of: rotation of the one or more cams about a rotation axis; and motion of the one or more cams along the first direction.
2. The expandable intervertebral implant of claim 1, further comprising a second slider threadably coupled to the first slider, wherein: the second slider is moveable along the first direction; the second slider is moveable independent of the first slider; and movement of the second slider results in at least one of: rotation of the one or more cams about the rotation axis; and motion of the one or more cams along the first direction.
3. The expandable intervertebral implant of claim 2, further comprising a first screw, extending along an anterior-posterior direction, configured to rotatably engage the first slider, wherein: the first direction comprises the anterior-posterior direction; and rotation of the first screw results in movement of the first slider in the anterior-posterior direction.
4. The expandable intervertebral implant of claim 3, wherein the first screw is captive relative to the second slider.
5. The expandable intervertebral implant of claim 3, further comprising a second screw, extending along the anterior-posterior direction, configured to rotatably engage the second slider; wherein the second screw is configured to rotate to urge movement of the second slider in the anterior-posterior direction.
6. The expandable intervertebral implant of claim 5, wherein the second screw is captive relative to the superior endplate.
7. The expandable intervertebral implant of claim 1, wherein the one or more cams are configured to be rotatable and translatable relative to the superior endplate and the inferior endplate.
8. An expandable intervertebral implant comprising: a superior endplate configured to engage a superior vertebra; an inferior endplate configured to engage an inferior vertebra, wherein the inferior endplate is spaced apart from the superior endplate by a gap; one or more cams configured to be translatable relative to the superior endplate and the inferior endplate; a first screw extending along an anterior-posterior direction; and a first slider configured to threadably receive the first screw; wherein: the first screw is configured to rotate to urge translation of the first slider in the anterior-posterior direction; the one or more cams are coupled to the first slider such that translation of the first slider in the anterior-posterior direction results in translation of the one or more cams; and the one or more cams are configured to urge angular expansion of the gap in

response to translation of the one or more cams.

9. The expandable intervertebral implant of claim 8, further comprising a second slider threadably coupled to the first slider and a second screw extending along the anterior-posterior direction, wherein: the second slider is moveable along the anterior-posterior direction; the second screw is configured to rotate to urge movement of the second slider in the anterior-posterior direction; and movement of the second slider urges linear expansion of the gap.

10. The expandable intervertebral implant of claim 9, wherein the first slider is moveable independent of the second slider.

11. The expandable intervertebral implant of claim 9, wherein movement of the second slider results in comparable movement of the first slider and movement of the first slider does not result in movement of the second slider.

12. The expandable intervertebral implant of claim 8, wherein the one or more cams are further configured to be rotatable relative to the superior endplate and the inferior endplate, wherein translation of the first slider in the anterior-posterior direction results in rotation of the one or more cams, and the one or more cams are configured to urge angular expansion of the gap in response to rotation of the one or more cams.

13. The expandable intervertebral implant of claim 8, further comprising: an anterior height comprising a first distance, in a superior-inferior plane, between a superior anterior portion and an inferior anterior portion; and a posterior height comprising a second distance, in the superior-inferior plane, between a superior posterior portion and an inferior posterior portion; wherein, in response to rotation of the one or more cams about a medial-lateral axis, the one or more cams are configured to urge angular expansion of the gap by increasing the anterior height more than the posterior height.

14. The expandable intervertebral implant of claim 8, wherein the one or more cams are received within one or more cam slots, and the one or more cam slots are configured to urge rotation of the one or more cams relative to the first slider in response to translation of the one or more cams within the one or more cam slots.

15. An expandable intervertebral implant comprising: a superior endplate configured to engage a superior vertebra; an inferior endplate configured to engage an inferior vertebra, wherein the inferior endplate is spaced apart from the superior endplate by a gap such that the expandable intervertebral implant comprises: an anterior height comprising a first distance, in a superior-inferior plane, between a superior anterior portion of the superior endplate and an inferior anterior portion of the inferior endplate; and a posterior height comprising a second distance, in the superior-inferior plane, between a superior posterior portion and an inferior posterior portion; and a first cam configured to rotate about a medial-lateral axis such that the gap expands linearly and angularly by: increasing the posterior height; and increasing the anterior height more than the posterior height.

16. The expandable intervertebral implant of claim 15, further comprising a first slider coupled to the first cam, wherein: the first slider is moveable along an anterior-posterior direction; and movement of the first slider results in at least one of: rotation of the first cam about the medial-lateral axis; and motion of the first cam along the anterior-posterior direction.

17. The expandable intervertebral implant of claim 16, further comprising a first screw, extending along the anterior-posterior direction, configured to rotatably engage the first slider, wherein the first screw is configured to rotate to urge movement of the first slider in the anterior-posterior direction.

18. The expandable intervertebral implant of claim 15, wherein the first cam is translatable in an anterior-posterior direction, and, in response to translation of the first cam, the first cam is configured to urge angular expansion of the gap by increasing the anterior height more than the posterior height.

19. The expandable intervertebral implant of claim 18, wherein the first cam is received within a

cam slot, and the cam slot is configured to urge rotation of the first cam relative to the cam slot in response to translation of the first cam within the cam slot.

20. The expandable intervertebral implant of claim 15, further comprising a second slider moveable along a first direction to urge linear expansion of the gap.
